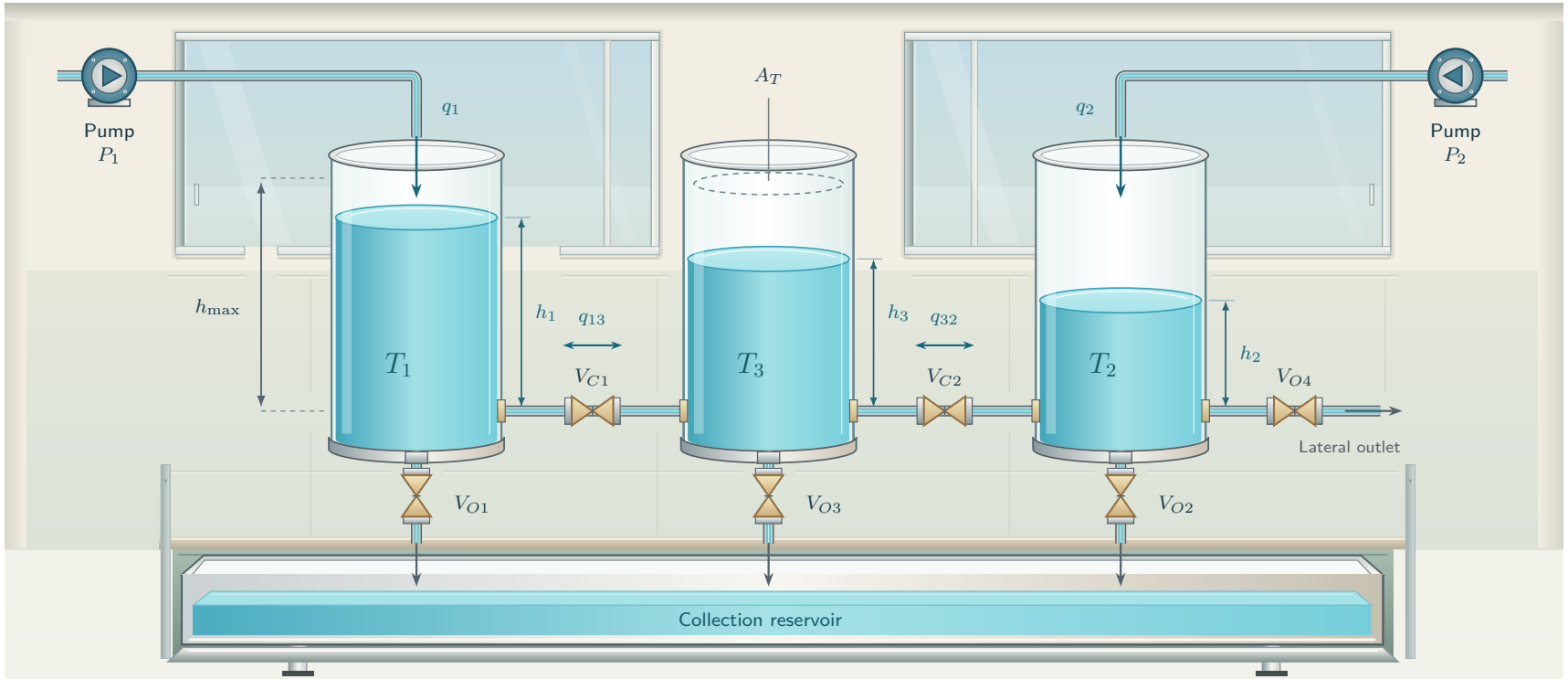
# Three interconnected tanks

Paper section 3.7: *Three-tank-system*

## Coupled free-surface tanks, independent pumps and selectable outlets




Illustrative hydraulic bench; not to scale. Outlet arrows show connections, not valve opening.

### Pump inputs and geometry

Independent inputs $q_1$ and $q_2$:
$0 \le q_1, q_2 \le 10^{-4}\ \mathrm{m^3\,s^{-1}}$.

Equal tank areas:
$A_T = 0.0154\ \mathrm{m^2}$.

Level limit: $h_{\max} = 0.62\ \mathrm{m}$.
State order: $[h_1, h_2, h_3]^\mathsf{T}$.

### Valve network and model scope

$V_{C1}$ couples $T_1$ and $T_3$.
$V_{C2}$ couples $T_3$ and $T_2$.

$V_{O1}$, $V_{O3}$, $V_{O2}$: bottom drains.
$V_{O4}$: lateral outlet from $T_2$.

**Nominal model: outlets closed.**
All outlet flows are zero. Opening an outlet requires a separately declared discharge extension.

### Signed hydraulic coupling

$$q_{ij} = a\mu\ \mathrm{sgn}(h_i - h_j)\sqrt{2g|h_i - h_j|}.$$

$a = 0.5\times10^{-4}\ \mathrm{m^2}$, $\mu = 0.48$.
Positive directions: $1 \to 3$ and $3 \to 2$.
Flow reverses when the head difference changes sign.

### Nominal level balances

$$A_T\dot{h}_1 = q_1 - q_{13}$$
$$A_T\dot{h}_3 = q_{13} - q_{32}$$
$$A_T\dot{h}_2 = q_2 + q_{32}$$

Time in seconds; levels in metres.

### Conservation and operating limit

$$A_T\frac{d}{dt}(h_1 + h_2 + h_3) = q_1 + q_2.$$

With closed outlets, positive net pumping increases total volume. Steady-flow tracking needs an explicit outlet model.

Scope: three-state nominal model, Eq. (36); complete valve topology follows Fig. 7. The collection basin is not an additional modeled state. Vessel levels, materials and bench details are illustrative; no measured geometry is inferred. Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.